Bitcoin Transaction Trace: DarkSide / Colonial Pipeline Ransom Proceeds

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Summary

Using publicly available blockchain explorers (mempool.space, WalletExplorer.com), I conducted a manual, bidirectional on-chain trace beginning from a Bitcoin address publicly documented in connection with the May 2021 DarkSide ransomware attack on Colonial Pipeline. The trace followed funds through five wallet hops, identified a previously unexamined second funding stream that converged with the original trail, and surfaced a classic layering/structuring pattern culminating in a partial cash-out at a KYC-regulated exchange.

Methodology

Forward- and backward-traced the transaction graph from the seed address; applied common-input-ownership clustering (via WalletExplorer's automated wallet grouping) to link addresses to controlling entities; logged dormancy intervals, consolidation events, and fan-out structure at each hop; cross-referenced tool-level service tags (e.g., named exchanges, CoinJoin participation) against each destination address.

Trace Findings

Hop	Wallet	Activity	Window	Dormancy
Origin	DarkSide Affiliate	Funds linked to May 2021 Colonial Pipeline ransom (FBI seizure-adjacent address)	Jun 2021	—
1	b6efa48710	Received 5.904 BTC; held in full; sent in full	Jun 2021 → May 2022	~11 months
2	81594aed41	Received full balance; held; sent in full	May 2022 → Sep 2023	~16 months
3	d28389097a	Consolidated with unrelated 1.631 BTC inflow; fanned out to ~65 addresses across 2 txns, incl. a CoinJoin-tagged address	Sep 21, 2023	13 min hold
3a	da3c47c7b5	Received fan-out share (0.911 BTC); fanned out again to ~30 addresses, incl. Kraken.com deposit (0.150 BTC)	Sep 21, 2023	18 min hold

Red-Flag Indicators Identified

Indicator	Evidence
Disposable single-hop wallets	Each wallet received one full deposit, sat dormant for months, then emptied to exactly 0.00000000 in one withdrawal — inconsistent with ordinary personal wallet use.
Multi-source consolidation	A second, independently-sourced 1.631 BTC stream (origin: wallet 30d740f334, itself preceded by a small “test transaction”) converged with the traced funds at a single timestamp, indicating common control.
Structuring / fan-out	A single transaction split its balance across ~30–65 irregular, non-round outputs — a recognized technique for fragmenting funds to resist manual tracing.
Confirmed mixing activity	One fan-out destination was independently flagged by the analysis tool as a CoinJoin-participating address.
Escalating dormancy	Hold periods increased from ~11 to ~16 months across successive hops, consistent with increasing caution — possibly in response to public reporting on the underlying case.
Named exchange off-ramp	A portion of traced funds (0.150 BTC) was deposited to Kraken.com, a KYC-regulated exchange — the point at which on-chain pseudonymity ends and account-level identity (subpoena-accessible) begins.

Conclusion & Limitations

The traced wallet sequence exhibits a coherent, multi-stage layering pattern consistent with deliberate money laundering, anchored to a publicly documented criminal proceeds case. The Kraken deposit represents an actionable off-ramp where subpoena-based identity attribution becomes possible — it is not itself evidence of wrongdoing, but the point where on-chain analysis hands off to legal process. A downstream high-volume wallet (2.25M+ transactions, ~16,800 BTC balance) showed a structurally similar batching pattern but could not be conclusively distinguished, from on-chain data alone, between a mixing service and a legitimate high-volume payment processor or exchange treasury wallet — illustrating the limits of transaction-graph analysis without external attribution data (e.g., sanctions lists, abuse databases, or proprietary risk-scoring tools).